

02_how_antonelli_wins_races

June 17, 2026

1 How Antonelli Keeps Winning — Race Results (Chapter 2)

Chapter 1 showed Antonelli usually out-qualifies his teammate — that’s *getting to the front*. This chapter asks the next question: **how do those grids become wins?** Through Monaco he has won **5 of the last 6** (China, Japan, Miami, Canada, Monaco), losing only the Australia opener.

Same control as Chapter 1 — his teammate **Russell** (same car) — plus, for race pace, the **actual P2 finisher** of each race as a field reference (it varies by race). Sign convention throughout: **positive = Antonelli better** (more positions gained, faster pace, lower degradation). Clean lap = green-flag racing lap (`TrackStatus == '1'`, not lap 1, not an in/out/pit lap).

```
[1]: %matplotlib inline
import sys; sys.path.insert(0, '..')
import warnings; warnings.filterwarnings('ignore')
import pandas as pd
import matplotlib.pyplot as plt

from src.loaders import setup_cache
from src import race
from src.season import RACES    # single source of truth (through Monaco)
from src.plotting import (plot_start_conversion, plot_stint_pace,
                           plot_gap_trace, plot_tire_deg)

setup_cache('../fastf1_cache')
RACES
```

```
[1]: ['Australia', 'China', 'Japan', 'Miami', 'Canada', 'Monaco', 'Barcelona']
```

1.1 The race record

Grid, lap-1 position, positions gained off the line, and finish for Antonelli at every round.

```
[2]: # Keyed by the RACES input string so the per-race P2 field reference can be
# looked up exactly in the pace cell below (not by fuzzy EventName match).
starts = {r: race.start_summary(2026, r) for r in RACES}
start = pd.concat(starts.values(), ignore_index=True)
start[start['driver'] == 'ANT'][['race', 'grid', 'lap1_pos', 'positions_gained', 'finish', 'status']]
```

core INFO Loading data for Australian Grand Prix - Race [v3.8.1]

```

req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
req          INFO      Using cached data for track_status_data
req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         WARNING    No lap data for driver 81
core         WARNING    Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 81)
core         INFO      Finished loading data for 22 drivers: ['63', '12', '16',
'44', '1', '3', '87', '41', '5', '10', '31', '23', '30', '43', '55', '11', '18',
'14', '77', '6', '81', '27']
core         INFO      Loading data for Chinese Grand Prix - Race [v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
req          INFO      Using cached data for track_status_data
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req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         WARNING    Driver 12 completed the race distance 00:00.022000
before the recorded end of the session.
core         INFO      Finished loading data for 22 drivers: ['12', '63', '44',
'16', '87', '10', '30', '6', '55', '43', '27', '41', '77', '31', '11', '3',
'14', '18', '81', '1', '5', '23']
core         INFO      Loading data for Japanese Grand Prix - Race [v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
req          INFO      Using cached data for track_status_data

```

```

req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         INFO      Finished loading data for 22 drivers: ['12', '81', '16',
'63', '1', '44', '10', '3', '30', '31', '27', '6', '5', '41', '55', '43', '11',
'14', '77', '23', '18', '87']
core         INFO      Loading data for Miami Grand Prix - Race [v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
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req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         WARNING    Driver 12 completed the race distance 00:00.027000
before the recorded end of the session.
core         INFO      Finished loading data for 22 drivers: ['12', '1', '81',
'63', '3', '44', '43', '16', '55', '23', '87', '5', '31', '41', '14', '11',
'18', '77', '27', '30', '10', '6']
core         INFO      Loading data for Canadian Grand Prix - Race [v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
req          INFO      Using cached data for track_status_data
req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         INFO      Finished loading data for 22 drivers: ['12', '44', '3',
'16', '6', '43', '30', '10', '55', '87', '81', '27', '5', '31', '18', '77',
'11', '1', '63', '14', '23', '41']
core         INFO      Loading data for Monaco Grand Prix - Race [v3.8.1]
req          INFO      Using cached data for session_info

```

```

req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
req          INFO      Using cached data for track_status_data
req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         INFO      Finished loading data for 22 drivers: ['12', '44', '10',
'6', '81', '30', '41', '23', '31', '14', '5', '63', '27', '43', '11', '55',
'16', '18', '1', '87', '77', '3']
core         INFO      Loading data for Barcelona Grand Prix - Race [v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      No cached data found for session_status_data. Loading
data...
_api         INFO      Fetching session status data...
req          INFO      Data has been written to cache!
req          INFO      No cached data found for lap_count. Loading data...
_api         INFO      Fetching lap count data...
req          INFO      Data has been written to cache!
req          INFO      No cached data found for track_status_data. Loading
data...
_api         INFO      Fetching track status data...
req          INFO      Data has been written to cache!
req          INFO      No cached data found for _extended_timing_data. Loading
data...
_api         INFO      Fetching timing data...
_api         INFO      Parsing timing data...
req          INFO      Data has been written to cache!
req          INFO      No cached data found for timing_app_data. Loading
data...
_api         INFO      Fetching timing app data...
req          INFO      Data has been written to cache!
core         INFO      Processing timing data...

```

```
core          WARNING      Fixed incorrect tyre stint information for driver '87'

core          INFO         Finished loading data for 22 drivers: ['44', '63', '1',
'3', '81', '6', '10', '30', '41', '43', '5', '55', '31', '11', '16', '12', '87',
'23', '14', '27', '77', '18']
```

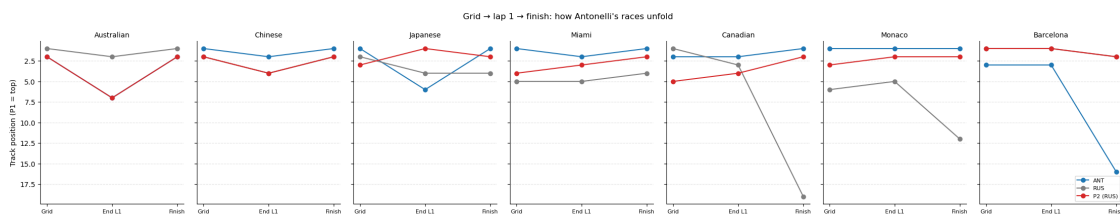
```
[2]:
```

	race	grid	lap1_pos	positions_gained	finish	status
0	Australian Grand Prix	2.0	7.0	-5.0	2.0	Finished
3	Chinese Grand Prix	1.0	2.0	-1.0	1.0	Finished
6	Japanese Grand Prix	1.0	6.0	-5.0	1.0	Finished
9	Miami Grand Prix	1.0	2.0	-1.0	1.0	Finished
12	Canadian Grand Prix	2.0	2.0	0.0	1.0	Finished
15	Monaco Grand Prix	1.0	1.0	0.0	1.0	Finished
18	Barcelona Grand Prix	3.0	3.0	0.0	16.0	Retired

1.2 Chapter A — Start & lap 1

Where each race is won or lost off the line and through the first lap.

```
[3]: fig = plot_start_conversion(start, save_path='../figures/start_conversion.png');
      ↪ plt.show()
```



Canada (R5) is the centerpiece: Antonelli started P2, led by lap 2, and won after Russell (on pole) retired — a win that was *converted* at the start and then *inherited* at the front. Monaco (R6) is the clean opposite: pole to flag-to-flag victory, no inheritance. Australia is the honest counter-case — P2 on the grid, P2 at the flag. That's a *finishing*-position loss, not a start failure: the front-row pace simply wasn't enough to pass Russell that day.

1.3 Chapter B — Race pace & control

Median green-flag lap per stint (teammate + the per-race P2 finisher), then a per-lap gap trace.

```
[4]: # Teammate pace + the per-race P2 finisher as field reference. The P2 code is
# read from each race's start summary (exact lookup, no fuzzy matching).
pace_frames = []
for r in RACES:
    p2_rows = starts[r][starts[r]['driver'] == 'P2']
    field = [p2_rows['code'].iloc[0]] if len(p2_rows) else []
    drivers = list(dict.fromkeys(['ANT', 'RUS'] + field)) # dedup if P2 is ANT/
    ↪RUS
```

```

    pace_frames.append(race.stint_pace(2026, r, drivers))
pace = pd.concat(pace_frames, ignore_index=True)
fig = plot_stint_pace(pace, save_path='../figures/stint_pace.png'); plt.show()

```

```

core          INFO      Loading data for Australian Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
req           INFO      Using cached data for driver_info
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'14', '77', '6', '81', '27']
core          INFO      Loading data for Chinese Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
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req           INFO      Using cached data for session_status_data
req           INFO      Using cached data for lap_count
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'14', '18', '81', '1', '5', '23']
core          INFO      Loading data for Japanese Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info

```

```

req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
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req          INFO      Using cached data for session_status_data
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req          INFO      Using cached data for session_info
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req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...

```

```

core          INFO      Finished loading data for 22 drivers: ['12', '44', '3',
'16', '6', '43', '30', '10', '55', '87', '81', '27', '5', '31', '18', '77',
'11', '1', '63', '14', '23', '41']

core          INFO      Loading data for Monaco Grand Prix - Race [v3.8.1]

req           INFO      Using cached data for session_info

req           INFO      Using cached data for driver_info

req           INFO      Using cached data for session_status_data

req           INFO      Using cached data for lap_count

req           INFO      Using cached data for track_status_data

req           INFO      Using cached data for _extended_timing_data

req           INFO      Using cached data for timing_app_data

core          INFO      Processing timing data...

core          INFO      Finished loading data for 22 drivers: ['12', '44', '10',
'6', '81', '30', '41', '23', '31', '14', '5', '63', '27', '43', '11', '55',
'16', '18', '1', '87', '77', '3']

core          INFO      Loading data for Barcelona Grand Prix - Race [v3.8.1]

req           INFO      Using cached data for session_info

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req           INFO      Using cached data for lap_count

req           INFO      Using cached data for track_status_data

req           INFO      Using cached data for _extended_timing_data

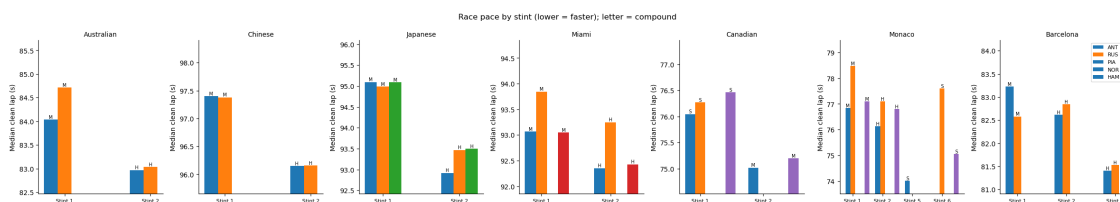
req           INFO      Using cached data for timing_app_data

core          INFO      Processing timing data...

core          WARNING    Fixed incorrect tyre stint information for driver '87'

core          INFO      Finished loading data for 22 drivers: ['44', '63', '1',
'3', '81', '6', '10', '30', '41', '43', '5', '55', '31', '11', '16', '12', '87',
'23', '14', '27', '77', '18']

```



Compare bars **within** a race and only across **matching compounds** (the letter on each bar). These are raw medians: they are **not** fuel-corrected, and stint timing and traffic differ between drivers — so read them as directional, not exact.

```
[5]: # Gap trace on the wins, where 'control' is the story. Negative = Antonelli
# leading (gap ahead of P2); positive = chasing (gap behind the leader).
gap = pd.concat([race.gap_to_rival(2026, r, 'ANT') for r in
↳ ['China', 'Japan', 'Miami', 'Canada', 'Monaco']],
                ignore_index=True)
fig = plot_gap_trace(gap, save_path='../figures/gap_trace.png'); plt.show()
```

```
core          INFO      Loading data for Chinese Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
req           INFO      Using cached data for driver_info
req           INFO      Using cached data for session_status_data
req           INFO      Using cached data for lap_count
req           INFO      Using cached data for track_status_data
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'16', '87', '10', '30', '6', '55', '43', '27', '41', '77', '31', '11', '3',
'14', '18', '81', '1', '5', '23']
core          INFO      Loading data for Japanese Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
req           INFO      Using cached data for driver_info
req           INFO      Using cached data for session_status_data
req           INFO      Using cached data for lap_count
req           INFO      Using cached data for track_status_data
req           INFO      Using cached data for _extended_timing_data
req           INFO      Using cached data for timing_app_data
core          INFO      Processing timing data...
core          INFO      Finished loading data for 22 drivers: ['12', '81', '16',
'63', '1', '44', '10', '3', '30', '31', '27', '6', '5', '41', '55', '43', '11',
'14', '77', '23', '18', '87']
```

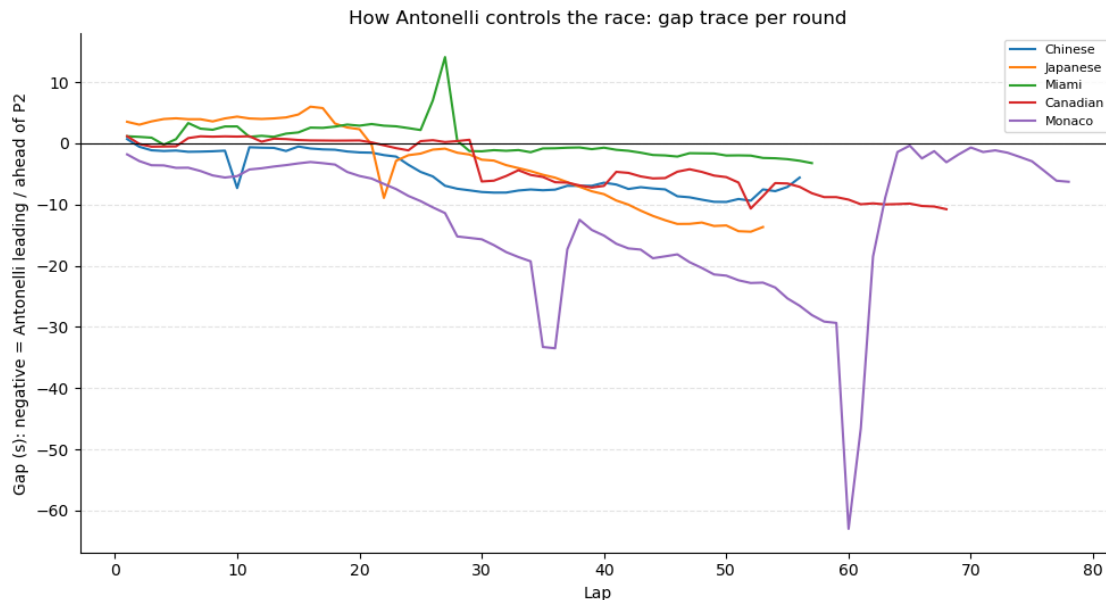
```

core          INFO      Loading data for Miami Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
req           INFO      Using cached data for driver_info
req           INFO      Using cached data for session_status_data
req           INFO      Using cached data for lap_count
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req           INFO      Using cached data for _extended_timing_data
req           INFO      Using cached data for timing_app_data
core          INFO      Processing timing data...
core          WARNING    Driver 12 completed the race distance 00:00.027000
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'18', '77', '27', '30', '10', '6']
core          INFO      Loading data for Canadian Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
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'16', '6', '43', '30', '10', '55', '87', '81', '27', '5', '31', '18', '77',
'11', '1', '63', '14', '23', '41']
core          INFO      Loading data for Monaco Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
req           INFO      Using cached data for driver_info
req           INFO      Using cached data for session_status_data
req           INFO      Using cached data for lap_count
req           INFO      Using cached data for track_status_data
req           INFO      Using cached data for _extended_timing_data
req           INFO      Using cached data for timing_app_data

```

```
core          INFO      Processing timing data...

core          INFO      Finished loading data for 22 drivers: ['12', '44', '10',
'6', '81', '30', '41', '23', '31', '14', '5', '63', '27', '43', '11', '55',
'16', '18', '1', '87', '77', '3']
```



At China, Japan, Miami and Monaco the trace sits below zero and *pulls away* — Antonelli leads and extends. Canada is the exception: the lead came partly inherited (Russell's retirement), which the honest-accounting note below names plainly.

1.4 Chapter C — Tire degradation

Linear slope of clean-lap time vs tire age per stint (lower = better tire management). Stints with fewer than 5 clean laps report NaN rather than a noisy fitted number.

```
[6]: deg = pd.concat([race.tire_deg(2026, r, ['ANT', 'RUS']) for r in RACES],
    ignore_index=True)
fig = plot_tire_deg(deg, save_path='../figures/tire_deg.png'); plt.show()
deg # includes NaN small-n stints, shown rather than hidden
```

```
core          INFO      Loading data for Australian Grand Prix - Race [v3.8.1]

req           INFO      Using cached data for session_info

req           INFO      Using cached data for driver_info

req           INFO      Using cached data for session_status_data

req           INFO      Using cached data for lap_count

req           INFO      Using cached data for track_status_data
```

```

req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         WARNING    No lap data for driver 81
core         WARNING    Failed to perform lap accuracy check - all laps marked
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core         INFO      Finished loading data for 22 drivers: ['63', '12', '16',
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core         INFO      Loading data for Chinese Grand Prix - Race [v3.8.1]
req          INFO      Using cached data for session_info
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req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
req          INFO      Using cached data for track_status_data
req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         WARNING    Driver 12 completed the race distance 00:00.022000
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core         INFO      Finished loading data for 22 drivers: ['12', '63', '44',
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core         INFO      Loading data for Japanese Grand Prix - Race [v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for lap_count
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core         INFO      Processing timing data...
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'14', '77', '23', '18', '87']

```

```

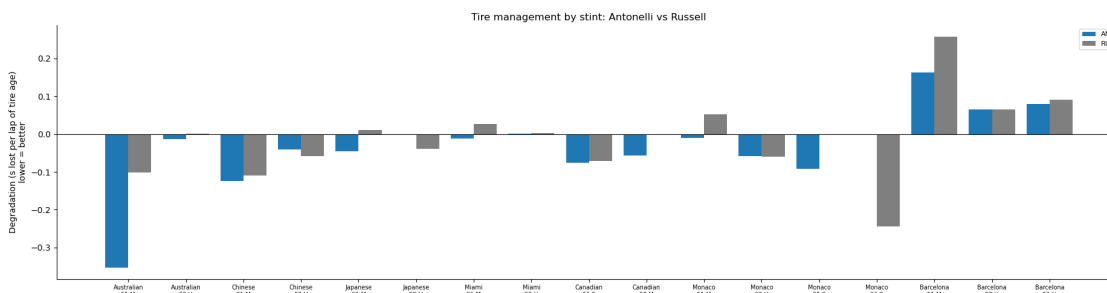
core          INFO      Loading data for Miami Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
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core          INFO      Loading data for Canadian Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
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core          INFO      Loading data for Monaco Grand Prix - Race [v3.8.1]
req           INFO      Using cached data for session_info
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```

```

core          INFO      Processing timing data...
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req           INFO      Using cached data for timing_app_data
core          INFO      Processing timing data...
core          WARNING    Fixed incorrect tyre stint information for driver '87'
core          INFO      Finished loading data for 22 drivers: ['44', '63', '1',
'3', '81', '6', '10', '30', '41', '43', '5', '55', '31', '11', '16', '12', '87',
'23', '14', '27', '77', '18']

```



```

[6]:
      race driver  stint compound  deg_slope_s_per_lap  n_clean
0  Australian Grand Prix  ANT      1  MEDIUM          -0.352550         9
1  Australian Grand Prix  ANT      2   HARD          -0.013633        40
2  Australian Grand Prix  RUS      1  MEDIUM          -0.101576        10
3  Australian Grand Prix  RUS      2   HARD           0.000661        40
4   Chinese Grand Prix  ANT      1  MEDIUM          -0.123893         8
5   Chinese Grand Prix  ANT      2   HARD          -0.040065        39
6   Chinese Grand Prix  RUS      1  MEDIUM          -0.109643         8
7   Chinese Grand Prix  RUS      2   HARD          -0.058989        39
8   Japanese Grand Prix  ANT      1  MEDIUM          -0.044729        20
9   Japanese Grand Prix  ANT      2   HARD          -0.000037        26
10  Japanese Grand Prix  RUS      1  MEDIUM           0.010025        19

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11	Japanese Grand Prix	RUS	2	HARD	-0.039286	26
12	Miami Grand Prix	ANT	1	MEDIUM	-0.011863	16
13	Miami Grand Prix	ANT	2	HARD	0.000217	28
14	Miami Grand Prix	RUS	1	MEDIUM	0.025932	10
15	Miami Grand Prix	RUS	2	HARD	0.002263	34
16	Canadian Grand Prix	ANT	1	SOFT	-0.076383	25
17	Canadian Grand Prix	ANT	2	MEDIUM	-0.057454	31
18	Canadian Grand Prix	RUS	1	SOFT	-0.071833	25
19	Monaco Grand Prix	ANT	1	MEDIUM	-0.010001	35
20	Monaco Grand Prix	ANT	2	HARD	-0.058074	21
21	Monaco Grand Prix	ANT	5	SOFT	-0.091429	7
22	Monaco Grand Prix	RUS	1	MEDIUM	0.051796	29
23	Monaco Grand Prix	RUS	2	HARD	-0.059137	26
24	Monaco Grand Prix	RUS	6	SOFT	-0.243700	5
25	Barcelona Grand Prix	ANT	1	MEDIUM	0.163178	12
26	Barcelona Grand Prix	ANT	2	HARD	0.064916	21
27	Barcelona Grand Prix	ANT	3	HARD	0.078997	20
28	Barcelona Grand Prix	RUS	1	MEDIUM	0.256794	10
29	Barcelona Grand Prix	RUS	2	HARD	0.065527	22
30	Barcelona Grand Prix	RUS	3	HARD	0.090809	22

Interpret only comparable **same-compound** stints. Fuel burn lowers lap times through a stint and partially offsets real degradation — that is **not** corrected for here, only named. Small-n stints are reported as NaN, never as zero.

1.5 Honest accounting

- **Canada’s win was partly inherited** — Russell led from pole and retired; Antonelli was P2 when it happened. Named, not hidden.
- **Six-race sample**, and the race mechanisms have even thinner data than the qualifying chapter. Directional, not conclusive.
- **Safety-car / VSC laps excluded everywhere** via the shared clean-lap filter.
- **No fuel-load, track-evolution, or strategy correction** — those factors are named as caveats, not modeled away.

1.6 What this chapter shows

Qualifying gets Antonelli to the front (Chapter 1); here we see he generally *converts* the start and *controls* the race when he leads — most cleanly at Monaco, most luckily at Canada. Tire management is comparable to Russell’s on like-for-like stints. The car is shared with Russell throughout, so what’s left is mostly the driver — with the caveats above. **Chapter 3** zooms out across seasons to ask whether his strong tracks are driver tracks or car tracks.